

Visual Studio Code Guide

Overview, Key Features & Essential Extensions for Modern Developers

What is Visual Studio Code?

Visual Studio Code (VS Code) is a lightweight, open-source, highly customizable code editor developed by Microsoft. Built on top of Electron and Node.js, it supports hundreds of programming languages out-of-the-box, providing fast startup times, integrated debugging, git version control, and an expansive extension marketplace.

Core Features of VS Code

- **Built-in Git Integration:** Review diffs, stage files, make commits, and manage branches directly from the editor sidebar.
- **IntelliSense:** Intelligent code completion based on variable types, function definitions, and imported modules.
- **Integrated Terminal:** Run scripts, CLI tools, and build commands without switching windows (`Ctrl + ``).
- **Multi-Language Support:** First-class support for JavaScript, TypeScript, Python, Java, C++, HTML/CSS, and Go.
- **Customizable Workspace:** Flexible panel layouts, customizable keybindings, themes, and workspace settings.

Essential Productivity Extensions

Extension Name	Category	Description & Purpose
Prettier	Code Formatting	An opinionated code formatter that enforces consistent code style across JS, TS, CSS, HTML, JSON, and Markdown automatically on save.
ESLint	Linting & Quality	Integrates ESLint into VS Code to analyze JavaScript and TypeScript code for errors, anti-patterns, and style violations in real time.
Live Server	Web Development	Launches a local development server with live reload feature for static & dynamic web pages directly from your workspace.
GitLens	Version Control	Supercharges VS Code's built-in Git capabilities with inline author blame, commit history navigation, file heatmaps, and repository insights.
Auto Rename Tag	HTML / XML	Automatically renames paired HTML/XML tags when you modify either the opening or closing tag, saving time and preventing syntax errors.

Extension Name	Category	Description & Purpose
Error Lens	Debugging	Enhances error highlighting by displaying diagnostic messages inline right next to the code line where the error occurs.

Top AI & Development Acceleration Extensions

Extension Name	Category	Description & Purpose
GitHub Copilot	AI Assistant	Provides real-time code completions, whole-function suggestions, unit test generation, and chat assistance directly inside your editor.
Path Intellisense	Autocompletion	Autocompletes local file paths in imports and relative path strings, preventing broken references across complex directory trees.
Docker	DevOps / Cloud	Adds commands and syntax highlighting for working with Dockerfiles, docker-compose.yml files, containers, and container registries.
Thunder Client	API Testing	A lightweight, postman-like REST API client extension built natively inside VS Code for rapid endpoint testing without external apps.
Material Icon Theme	UI & Theme	Applies distinct, visually intuitive icons to files and directories based on file extensions and project structure for improved file navigation.

Essential VS Code Keyboard Shortcuts

Ctrl + Shift + P (or Cmd + Shift + P): Open Command Palette	Ctrl + P (or Cmd + P): Quick Open File by Name
Ctrl + / (or Cmd + /): Toggle Line Comment	Alt + Up/Down (or Option + Up/Down): Move Line Up or Down
Ctrl + D (or Cmd + D): Select Next Occurrence of Current Word	Ctrl + `</b> (or <b>Cmd</b> + <b>`): Toggle Integrated Terminal